COMPUTER VISION PRACTICAL REPORT

Devalla Pranav

devallapranav192k5@gmail.com

AI/ML intern

Role: Computer Vision

Introduction

This report presents the implementation of various Computer Vision modules developed using Python, OpenCV, and MediaPipe. The programs mainly focus on human pose estimation, body segmentation, cloth masking, dense pose representation, and virtual try-on preprocessing techniques.

These implementations use webcam-based real-time processing to identify and analyze different human body regions. The modules are useful in applications such as virtual try-on systems, motion tracking, activity recognition, and AI-based human analysis.

---

1. AGNOSTIC-v3.2

Introduction

Agnostic representation is a preprocessing technique used in virtual try-on systems. The purpose of this module is to remove clothing appearance information while preserving the body posture and structure of the user.

The upper body region is detected using MediaPipe pose estimation and then masked using a gray overlay.

Algorithm

1. Capture webcam frame
2. Convert frame into RGB format
3. Detect body landmarks
4. Identify torso coordinates
5. Generate polygon mask
6. Apply gray masking

7. Display output

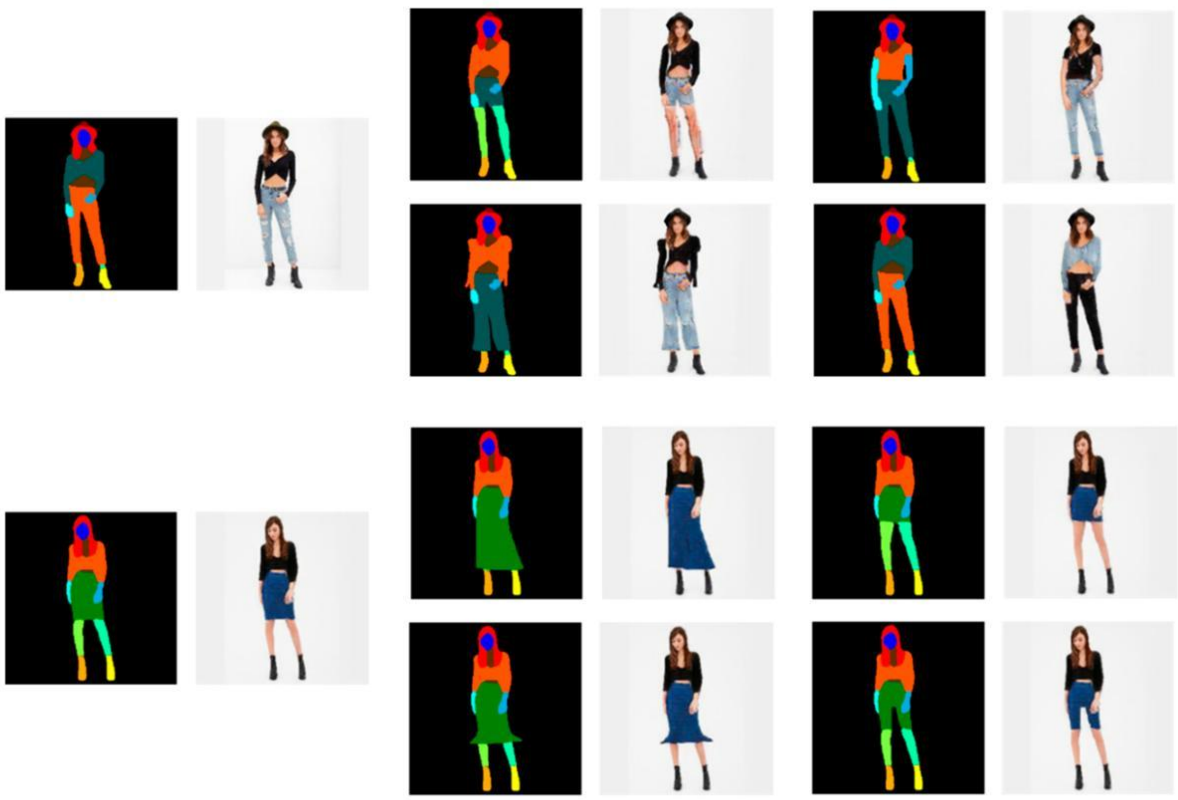
Technologies Used

- Python
- OpenCV
- MediaPipe
- NumPy

Applications

- Virtual Try-On Systems
- Human Body Preprocessing
- AI-based Garment Replacement
- Pose-Aware Image Generation

Output



(a) (b)

---

2. PARSE-AGNOSTIC-v3.2

Introduction

Parse-Agnostic converts the upper body region into segmentation masks. The implementation separates clothing regions from the remaining body structure and produces a binary mask output.

The segmented region appears white over a black background.

Algorithm

1. Capture webcam input
2. Detect body landmarks
3. Identify shoulder and torso region
4. Generate segmentation mask
5. Convert output into binary format
6. Display output

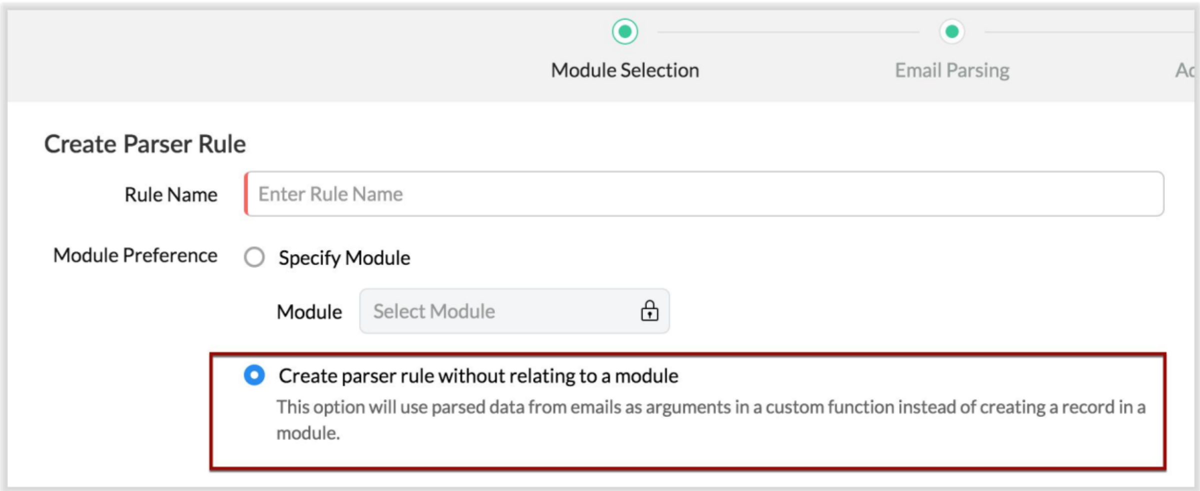
Advantages

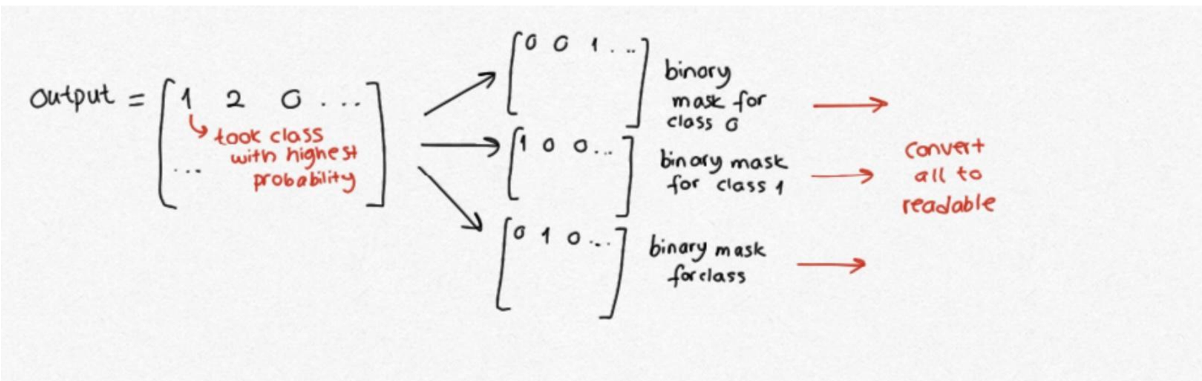
- Lightweight implementation
- Real-time processing
- Fast mask generation
- Useful preprocessing stage

Applications

- Cloth Segmentation
- Body Parsing
- AI Preprocessing
- Image Manipulation Systems

Output





---

3. PARSE-v3 BODY SEGMENTATION

Introduction

Parse-v3 performs semantic body segmentation by dividing the human body into different colored regions such as head, torso, arms, and lower body.

Each body part is represented using a different color for easy visualization.

Body Region Mapping

- Head → Blue
- Torso → Orange
- Arms → Cyan
- Lower Body → Green

Algorithm

1. Capture webcam frame
2. Detect body pose landmarks
3. Separate body regions
4. Assign colors to body parts
5. Generate segmented visualization
6. Display output

Advantages

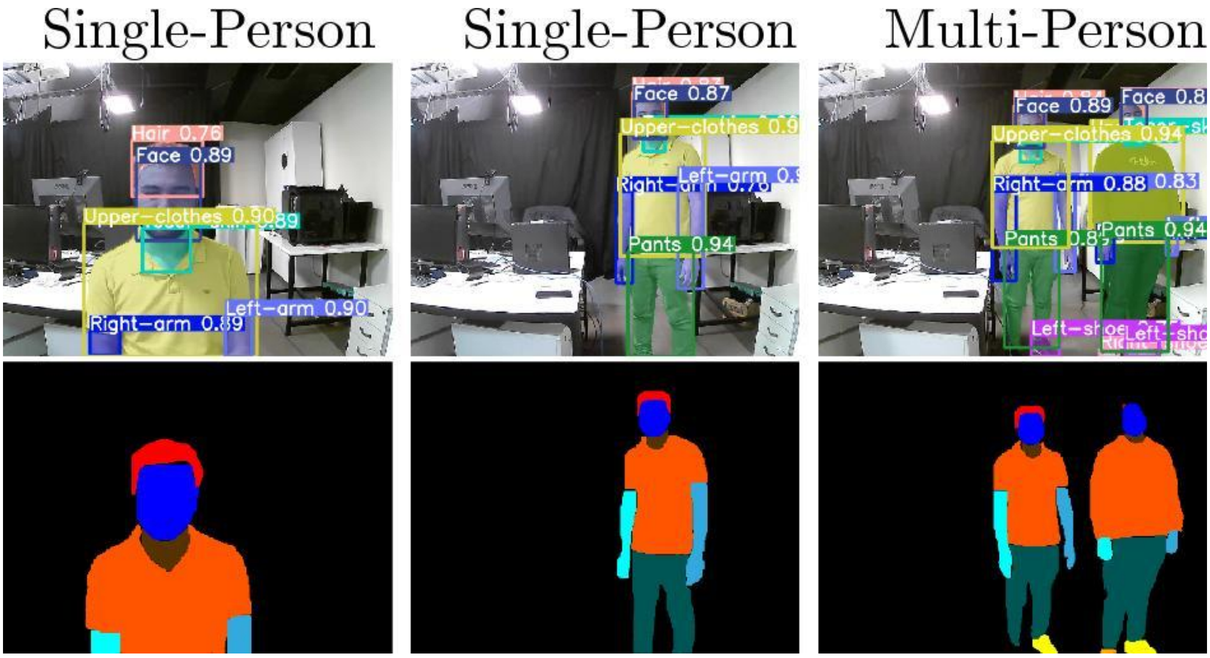
- Easy body analysis
- Better visualization
- Region identification
- Useful for semantic segmentation

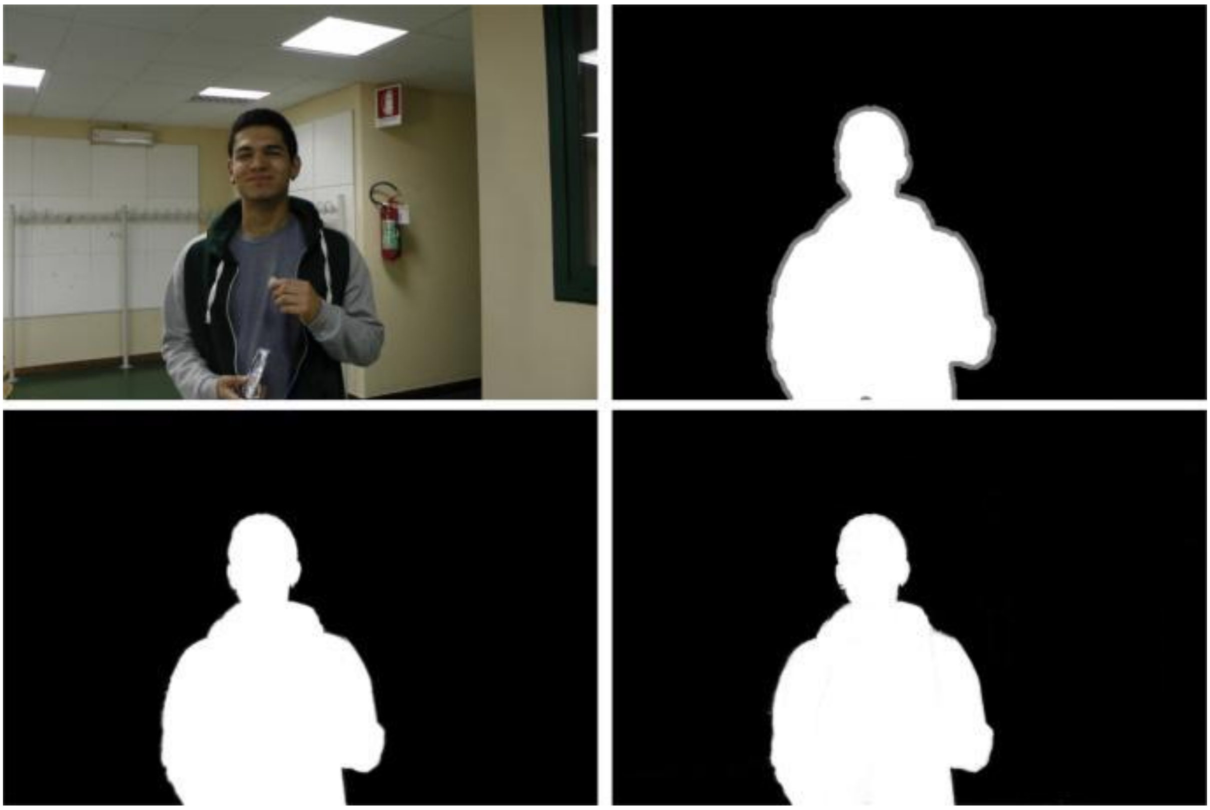
Applications

- Activity Recognition

- Human Tracking
- Motion Analysis
- Pose Detection

Output





4. DENSE POSE

Introduction

Dense Pose creates a dense body representation using color mapping techniques. Different body regions are visualized using distinct colors.

The module generates a simplified dense body representation using pose landmarks.

Working Principle

- Webcam captures image
- Pose landmarks are extracted
- Body regions are identified
- Dense body mapping is generated
- Color representation is displayed

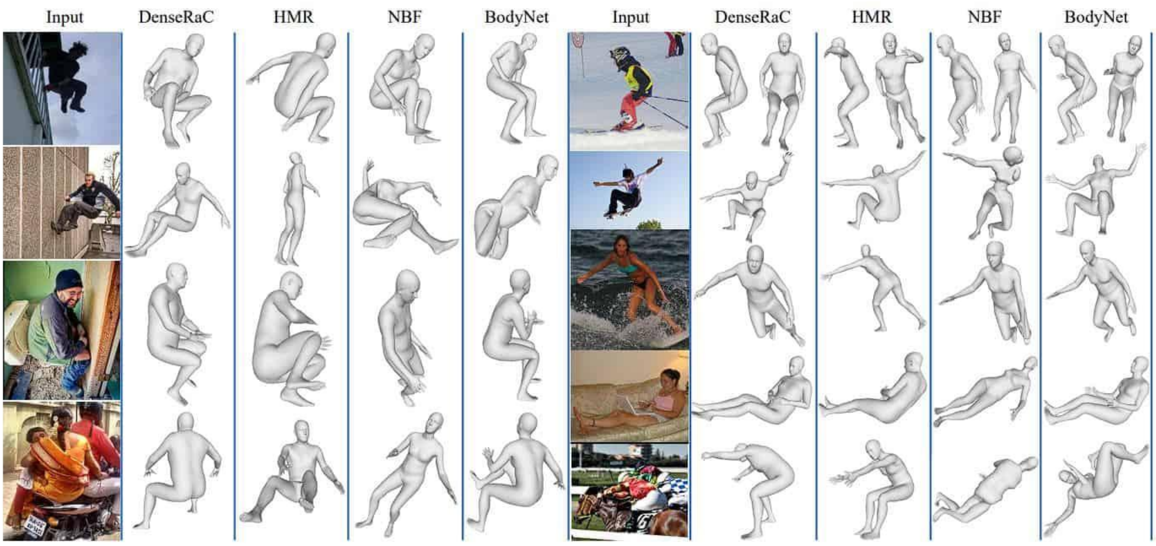
Advantages

- Real-time processing
- Dense body visualization
- Easy body structure analysis
- Lightweight implementation

Applications

- Animation Systems
- Human Body Representation
- Motion Tracking
- AI Vision Systems

Output



---

5. OPEN POSE

Introduction

Open Pose estimates human body joints and generates skeleton representations. Different body key points are detected and connected to form the human pose structure.

Working Principle

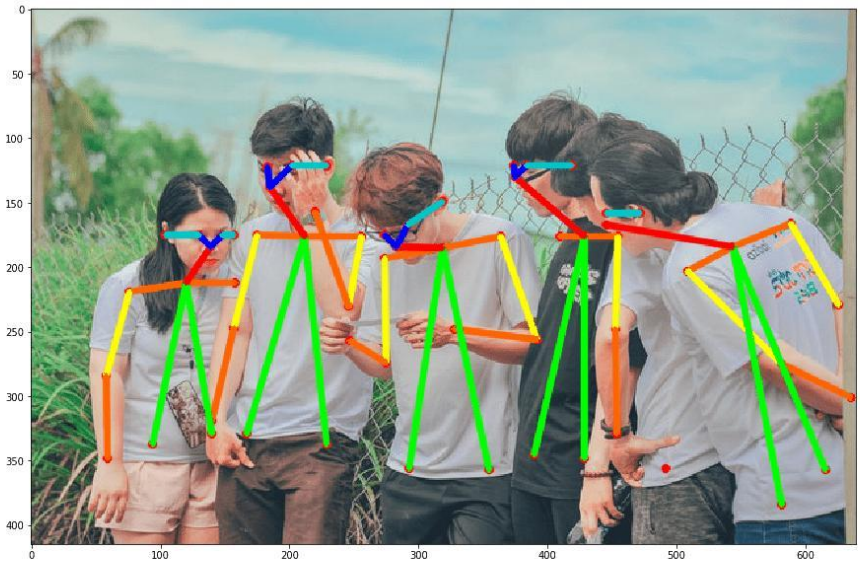
- Webcam captures image
- Body joints are detected
- Landmark coordinates are extracted
- Skeleton structure is generated
- Real-time pose output is displayed

Advantages

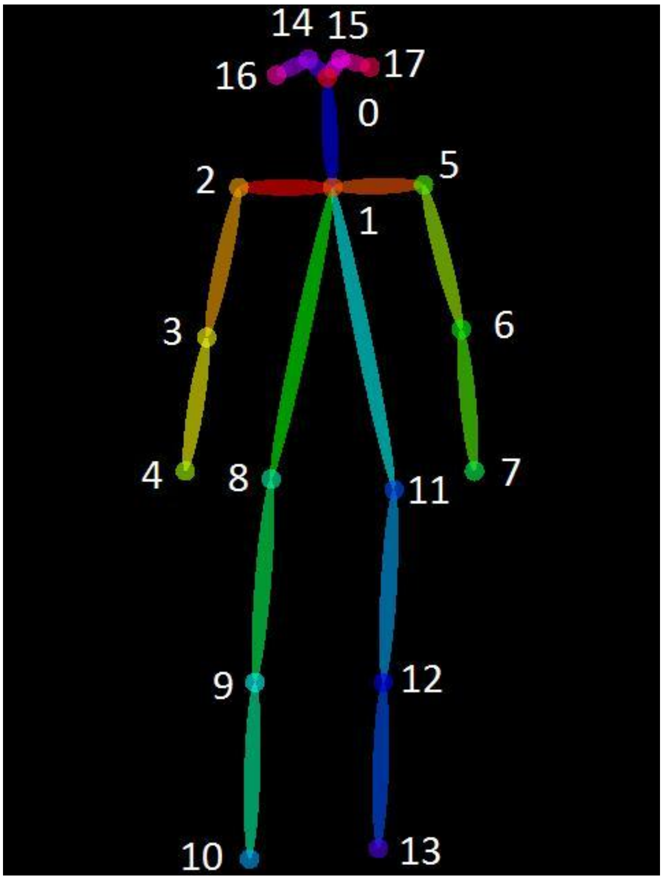
- Accurate pose tracking
- Lightweight implementation
- Easy visualization
- Real-time performance

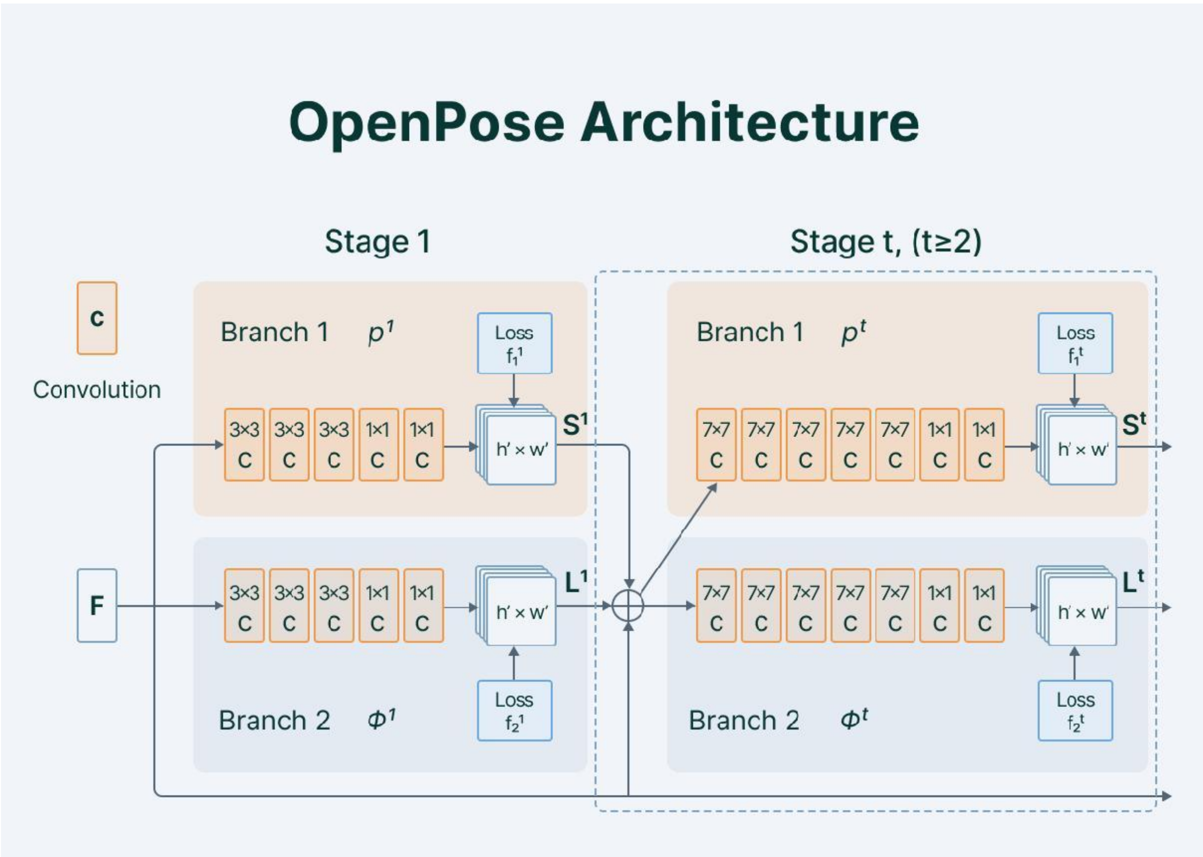
Applications

- Gesture Recognition
- Motion Tracking
- Gaming Systems
- Human Activity Detection



Output


14
15
16
17
0
2
5
1
3
6
4
8
11
7
9
12
10
13


OpenPose Architecture
Stage 1
Stage t, (t≥2)
C
Convolution
F
Branch 1 p¹
Branch 2 Φ¹
Branch 1 pᵗ
Branch 2 Φᵗ
Loss f₁¹
Loss f₂¹
Loss f₁ᵗ
Loss f₂ᵗ
3×3 C
1×1 C
7×7 C
h′ × w′
S¹
L¹
Sᵗ
Lᵗ

6. CLOTH MASK

Introduction

Cloth Mask extracts garment regions from webcam input and generates binary mask representations. The module separates the clothing region from the remaining background.

Working Principle

- Webcam captures image
- Pose landmarks are detected
- Garment region is identified
- Cloth extraction is performed
- Binary mask is generated

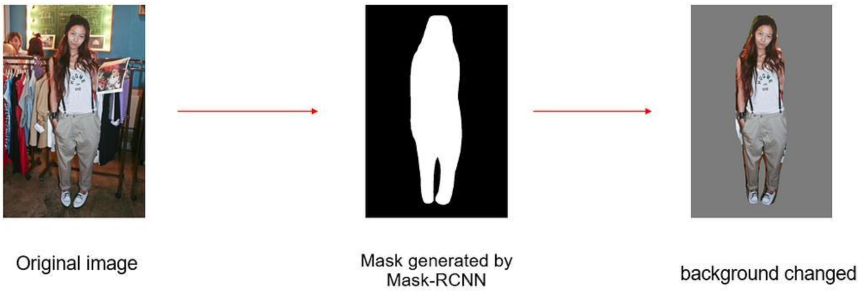
Advantages

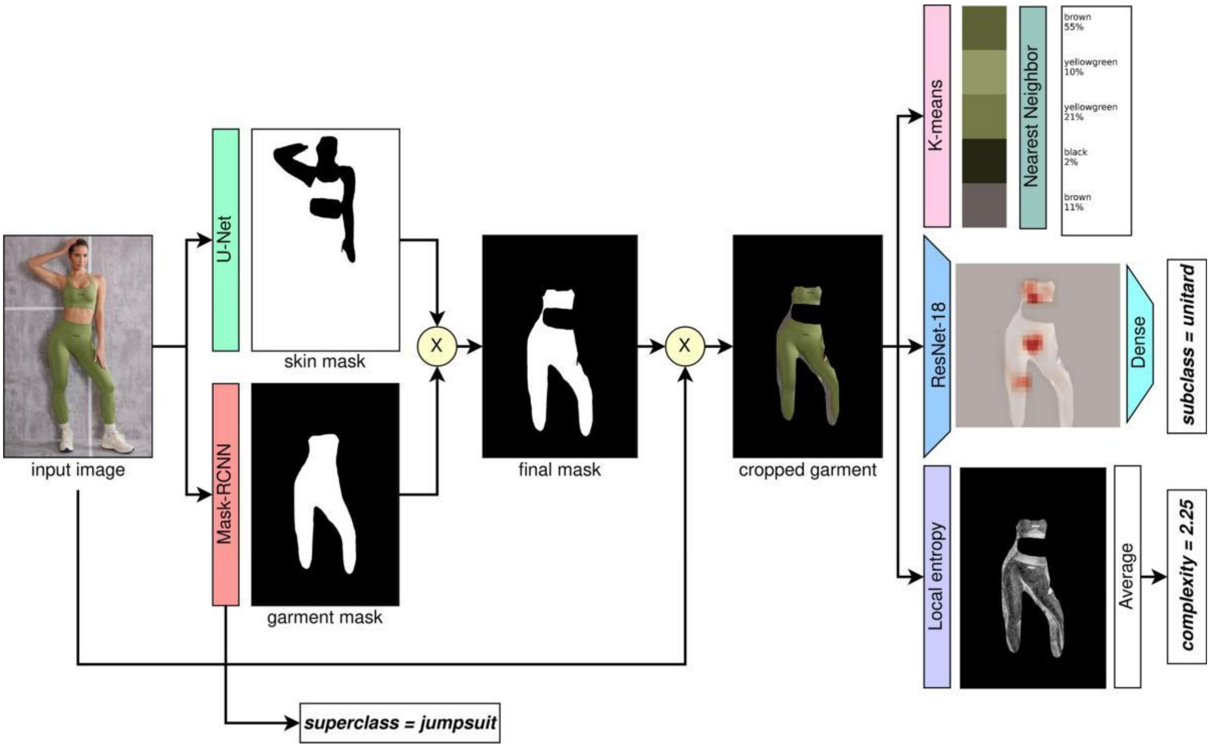
- Fast cloth extraction
- Efficient mask generation
- Real-time processing
- Useful preprocessing stage

Applications

- Virtual Try-On Systems
- Garment Segmentation
- Fashion AI Systems
- Cloth Extraction

Output





Common Libraries Used

Library	Purpose
Python	Core Programming Language
OpenCV	Image and Video Processing
MediaPipe	Pose Estimation
NumPy	Numerical Operations

Conclusion

The implemented modules demonstrate the importance of pose estimation and body segmentation in modern Computer Vision systems. By combining OpenCV with MediaPipe, efficient real-time systems can be developed for virtual try-on applications, motion tracking, human activity recognition, and intelligent visual processing.

These implementations provide a strong foundation for advanced AI-based vision applications and real-time human analysis systems.